

Multi-Asset Next-Day Return Prediction — Bitcoin, Gold, Silver

Final Year Project · Executive Summary for the Supervisor · Alyan Shahid
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This project is a machine-learning system that forecasts the **next trading day's return** of three assets — Bitcoin (BTC-USD), Gold (GC=F futures) and Silver (SI=F futures) — and presents the forecast, together with its measured reliability, in a web dashboard and a REST API. The forecast target is the next-day log return $y_t = \ln(P_{t+1}/P_t)$; the price shown to the user is recovered as $P_t \cdot \exp(\hat{y})$. Daily data from Yahoo Finance (2023-07-28 to 2026-07-28) is enriched with macro series (US Dollar Index, WTI crude, 10-year Treasury yield, S&P 500, VIX) and the crypto Fear & Greed index.

Methodology. A frozen chronological split (train $\leq$ 2025-09-10, validation $\leq$ 2026-02-17, test = 2026-02-18 $\rightarrow$ 2026-07-28) with no shuffling; a scaler fitted on training rows only; 23–24 stationary, backward-looking features; hyper-parameters and the served model chosen by 4-fold expanding-window walk-forward validation inside train+val; and a single evaluation of the untouched test set. Ten model families are compared on identical days: a random-walk baseline, historical mean, ARIMA, Ridge, Random Forest, LightGBM, CatBoost, GRU, LSTM and a stacked ensemble. Ten specific data-leakage checks are enforced by a 28-case test suite.

Result. The validation-selected model for all three assets is CatBoost (shallow, heavily regularised gradient-boosted trees). On the untouched test set its return-RMSE is within 0.3–1.5 % of the random-walk forecast (Diebold–Mariano $p = 0.15$ – 0.57), and its directional accuracy is 47.7–48.6 % (not significantly different from 50 %). **No model beats the random walk by a statistically significant margin**; the recurrent networks and ARIMA are significantly worse ($p \leq 0.05$). The tuned models converge to the unconditional drift, which is the expected outcome under weak-form market efficiency at a one-day horizon with ~ 700 – $1,000$ daily observations. The project therefore delivers a validated negative result, not a forecasting edge, and the application reports this honestly next to every forecast.

Improvement over the initial version. An audit of the original codebase found a price-reconstruction bug that corrupted every dollar metric, 30 % fabricated weekend rows for the metals, hyper-parameters selected on the test set, early stopping on training data, non-stationary price-level features, a served ensemble that predicted a constant, a broken inference path and misleading R^2 values (≈ 0.95 on price levels, which a random walk also achieves). All were fixed; the pipeline was re-run end-to-end and reproduces every reported number bit-for-bit.

Status. Working Streamlit dashboard (asset selection, indicator overlays, next-day forecast with uncertainty band, multi-model comparison, validation and test tables, figures) and FastAPI endpoints; reproducible pipeline (make pipeline); documentation for methodology, features, results, limitations and viva.

Key facts

Item	Final implementation
Assets / horizon	Bitcoin, Gold (GC=F), Silver (SI=F); next trading day
Target	log return $\ln(P_{t+1} / P_t)$; price = $P_t \cdot \exp(\hat{y})$
Data	Yahoo Finance daily OHLCV 2023-07-28 $\rightarrow$ 2026-07-28 (BTC 1,067 / metals 724 usable days) + DXY, WTI, 10-y yield, S&P; 500, VIX, Fear & Greed
Features	24 (BTC) / 23 (Gold) / 24 (Silver) stationary, backward-looking features; raw price levels never used as inputs
Split	train $\leq$ 2025-09-10 · validation $\leq$ 2026-02-17 · test 2026-02-18 $\rightarrow$ 2026-07-28 (161 / 111 / 111 days), chronological, evaluated once

Item	Final implementation
Validation	4-fold expanding-window walk-forward on train+val for tuning and model selection
Models	Naive random walk, historical mean, ARIMA, Ridge, Random Forest, LightGBM, CatBoost, GRU, LSTM, stacked ensemble
Served model	CatBoost (validation winner for all three assets; depth 3/5/3, lr 0.01)
Metrics	MAE/RMSE/MAPE (\$), RMSE/MAE/R ² (return), directional accuracy + binomial p, Diebold–Mariano vs random walk, strategy backtest
Stack	Python 3.9, pandas, scikit-learn, LightGBM, CatBoost, TensorFlow/Keras, statsmodels, Streamlit, FastAPI, Docker, pytest (28 cases), GitHub Actions

Headline test-set results

Asset	Model	MAE (\$)	RMSE (\$)	MAPE %	R ² (return)	Dir. acc. % (p)	DM p vs naive
Bitcoin	Naive (random walk)	1,093.48	1,421.12	1.58	-0.000	—	—
Bitcoin	CatBoost (served)	1,096.70	1,424.81	1.59	-0.005	47.8 (0.74)	0.57
Gold	Naive (random walk)	60.97	78.84	1.34	-0.010	—	—
Gold	CatBoost (served)	61.74	80.03	1.36	-0.041	47.7 (0.72)	0.15
Silver	Naive (random walk)	1.95	2.57	2.65	-0.004	—	—
Silver	CatBoost (served)	1.97	2.59	2.67	-0.017	48.6 (0.65)	0.27

Test period 2026-02-18 → 2026-07-28. Full 10-model tables: results/FINAL_RESULTS.md and the main report.

Reading the numbers

R² is reported on returns, where 0 means "as good as predicting the mean"; every model is within a few thousandths of 0 or below it. The original project reported R² ≈ 0.95 on price levels — a value the random walk also achieves, which is why it was misleading. The Diebold–Mariano p-values show that none of the served models' errors differ significantly from the random walk's; models that deviate more (ARIMA, GRU, LSTM) are significantly worse. The correct reading is that the best achievable one-day point forecast on this data is approximately the drift, and the system says so.

Current status: implementation complete; pipeline reproducible bit-for-bit; dashboard and API verified running for all three assets; documentation and viva notes prepared. **Recommended framing:** a leakage-audited forecasting benchmark and decision-support prototype with a validated negative result — not a market-beating predictor.